METRIC
DOD-P-63975 (AR)
AMENDMENT 7
29 November 1994
SUPERSEDING
AMENDMENT 6
15 February 1989

MILITARY SPECIFICATION

PRIMER, ELECTRIC, PA520

This amendment forms a part of DOD-P-63975 (AR), dated 14 May 1982, and is approved for use by the US Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

STANDARD, MILITARY:

Add "MIL-STD-129 - Marking for shipment and Storage".

Add "MIL-STD-1235 - Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes".

Delete "MIL-STD-331".

2.1.2, Drawings, Inspection Equipment:

Delete: "11825969 - Test Barrel, 30mm Mann (6.50)" and substitute "9390627 - Barrel, Test, 30mm (ADEN Barrel)"

* Add drawing: 9390748 - Barrel, Test: Gun, M230, 30mm (Progressive w/6.5 degree exit twist)

PAGE 3

* 3.6: Delete in its entirety and substitute the following:

"3.6 Continuity and resistance The primer circuit shall be continuous with an electrical resistance of the uncapped primer (dry) not less than 20 ohms (50 ohms desired) nor more than 1000 ohms. The electrical resistance of the capped primers shall not be less than 20 ohms (50 ohms desired) nor more than 1000 ohms."

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PAGE 4

3.9: Delete in its entirety and substitute the following:

"3.8 .....The primer shall have an all-fire voltage of 22 volts DC or less and no-fire voltage of 3.0 volts DC minimum. The primer performance shall not be degraded when fired at either 115 volts, 400 HZ or in the circuit of 3.9."

3 9: Delete in its entirety and substitute the following:

"3.9 Function time. Testing for function times of 3.9.1 and 3.9.2 shall be conducted ma 163 volt, 400 HZ half wave rectified circuit with a 100 ohm resistor in series with the primer pulse duration shall be 4.0 milliseconds."

Add 3.9.1 as follows:

"3.9.1 Electric primer time. The individual and mean plus three standard deviation primer time values shall not exceed .400 milliseconds."

3.10 should read "3.9.2".

PAGE 5

4.3.1:

Delete "125" and substitute "32".

Delete "Insulator" in its entirety.

Delete "1430" and substitute "1312".

Delete "MOD I".

PAGE 7

Table I:

Delete "MOD I".

Delete "125" and substitute "32" wherever it appears.

Delete "insulator" in its entirety.

Add: "Button-cup-insulator subassembly (reference 11825910) insulation strength. 7/, 32, 100%, 3.5, 4.5.1".

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PAGE 8

Change quantities to 32 pieces for: Examination for defects and delete "4.3.2.2" and add "4.4.2.4".

Firing Voltage: Change "600" to "550".

Delete "Insulation strength" in its entirety.

Table I in Requirements Paragraph column: Delete "3.9" and "3.10" and substitute "3.9.1" and "3.9.2" respectively.

PAGE 9

Note 5/: Delete "action time plus four" and substitute "action time plus three".

PAGE 10

4.4.1:

Delete last sentence of first paragraph. Delete items "a) thru e)"

4.4.2 a:

Add "or MIL-STD-1235" to the last sentence.

PAGE 11

4.4.2.1, Major 102: Add "/ORD-SIO-S314" to Visual.

PAGE 13

4.4.2.3, Major 102: Add "/ORD-SIO-5314" to Visual.

"3/ Rotation and tilting of primers under magnification may be required to view for presence of insulator."

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PAGE 14

4.4.2.4:

Add "11825901" to next higher assembly.

Critical 3: Delete "loss" and substitute "loose".

Critical 8: Delete "600" and substitute "550".

102: Add a "3/" after "missing, misplaced or protruding insulator" and 3/ should read:

PAGE 15

4.4.2.4:

Add "11825901" to next higher assembly.

Title: Delete footnote "1/"

Add new Major 106: "Primer time, n = 50, Acc-0, Rej-1, 3.9, 4.5.5".

Notes: Delete "8/" and substitute "2/"

Add: "Major 107, Action Time 3/, -50 ^{O}C, 100, 3.9.2, 4.5.6 +21 ^{O}C, 25, 3.9.2, 4.5.6
+71 ^{O}C, 25, 3.9.2, 4.5.6

3/ If the sample average action time plus three standard deviations exceeds 4 milliseconds or if an individual primer time of the test sample exceeds 4 milliseconds, the lot shall be rejected."

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PAGE 16

4.5.4: Delete "If the average ..." in its entirety and substitute:

"If the average critical voltage minus 2 standard deviations is less than 2 volts DC, the lot shall be retested. The retest shall consist of firing a random sample of 100 primers at 3 volts DC with an 8.2 ohm resistor placed in series with the primer. If any primers fire during this test, the lot shall be rejected."

4.5.4: Statement "e" should read "The primer firing device shall be set for a 4 millisecond pulse time."

PAGE 17

4.5.6: Delete in its entirety and substitute the following:

"4.5.6 Action time The test rounds, assembled in accordance with Drawing 11825947, shall be conditioned at -54°C, 21°C, and 71°C and fired from a test barrel in accordance with Drawing 9390748. The time between application of voltage to the primer and exit of the projectile from the muzzle shall be measured. The time shall comply with the requirement of paragraph 3.9.2. Any corrected action time of 2.40 milliseconds or less shall be regarded as an instrumentation error."

* 4.5.6: Delete the last sentence and substitute the following:

"Any corrected action time of 1.92 milliseconds or less shall be disregarded as instrumentation error."

4.5.7, line 11: Delete "busts of 25 rounds" and substitute "20 round minimum burst".

Delete "XM230E1" and substitute "M230" 4.5.8: Delete "in accordance with Table II".

4.5.9.1: Delete "4.5.6" and substitute "4.5.9".

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4.5.9.2: Delete first sentence and substitute "The test cartridge(s) shall be subjected to a board band random vibration profile in accordance with the requirements of MILSTD-810, Method 514/2, Procedure 1 Ca, "C" with the vibration spectrum per Figure 514.2-3 and 514.2.10 of MIL-STD-810."

Second sentence: Delete "AAH magazine" and substitute "M592 Ammo Can".

PAGE 18

4.5.9: Delete "MIL-STD-801" and substitute "MIL-STD-810".

4.5.10: Delete in its entirety.

4.5.11: Delete "4.5.11" and substitute "4.5.10".

5.1: The last sentence should read:

"The primers shall be packed in such manner as to be protected from moisture."

PAGE 19

* 6.3: Delete "DRDAR-QAF-I" and substitute "AMSTA-AR-QAC-C".

* 6.4: Delete "DRDAR-QAF-S" and substitute "AMSTA-AR-QAC-C".

PAGE 20

* 6.7: Delete "US Army Armament Research and Development Command (ARRADCOM)" and substitute "US Army Armament Research, Development and Engineering Center (ARDEC)"

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